

Nil Sahin

Website: <https://nsahin.github.io>

Work Experience

May 2023 – present **Senior Data Scientist at Recursion Pharmaceuticals**

- Led the full lifecycle development of core data pipelines and successfully refactored existing systems to standardize logic, increase data connectivity, and ensure compatibility with new platform architectures.
- Engineered and deployed scalable, automated workflows for benchmarking and relationship metrics, which significantly reduced model evaluation time and provided consistent, valuable data for drug discovery.
- Drove data product quality by leading critical updates to the benchmarks database and implementing new metrics that reduced the runtime of a core initiation workflow from multiple days to a few hours.
- Expanded influence by mentoring team members and interns, and actively contributed to multiple technical guilds (e.g., Observability, Model Evaluation) to foster best practices and knowledge sharing across the organization.

July 2021 – May 2023 **Computational Scientist at Cyclica, Inc.**

- Engineered a fully generalizable, semi-automated clustering module capable of grouping compounds using any feature set, which improves
- Automated the Drug Metabolism and Pharmacokinetics processing workflows to quantify experimental assays based on existing protocols in Excel sheets that reduced a couple of hours of manual labour to a few minutes
- Contributed to the Hit Expansion pipeline by building functions to access and integrate stock information from many chemical libraries

Education

2015 – 2021 **University of Toronto**, Toronto, Ontario, Canada
Ph.D., Department of Molecular Genetics, Faculty of Medicine
Co-supervisors: Dr. Brenda J. Andrews, Dr. Quaid Morris

2011 – 2015 **Sabanci University**, Istanbul, Turkey
BSc., Molecular Biology, Genetics and Bioengineering
Faculty of Engineering and Natural Sciences

2006 – 2011 **Robert College**, Istanbul, Turkey

Research Experience

2016 – 2021 **Andrews Laboratory, University of Toronto**
Supervisor: Dr. Brenda J. Andrews

An Atlas of Morphological Phenotypes Associated with Gene Perturbations

Developed image analysis pipelines to identify genetic regulators of cellular morphology in the context of genome-wide single gene perturbations in yeast by applying machine learning and computer vision strategies

2015, Oct. – Dec. **Morris Laboratory, University of Toronto**
Supervisor: Dr. Quaid Morris

Mutational Signature Changes during Tumour Evolution

Implemented bioinformatics algorithms to quantify mutational signatures of 600 tumours from whole genome sequencing; identified significant signature differences between tumour types; created an opportunity for a new PhD thesis for a new student after this 5-week rotation project

2013 – 2015 **Cokol Laboratory, Sabanci University**
Supervisor: Dr. Murat Cokol

Large-scale Experimental *E. coli* Drug Interactions Screen

Conducted drug interaction experiments among 25 antibiotics; analyzed results to find significantly synergistic drug pairs and relate them to mechanism of action of each antibiotic

2014, June – Aug. **Roth Laboratory, University of Toronto**

Supervisor: Dr. Frederick P. Roth

Drug Sensitivity Profiles of ABC Transporter deletion strains in *S. cerevisiae*

Conducted screen to identify relationship between ABC transporters and drug sensitivity by using 16 *S. cerevisiae* strains and measured sensitivity of strains to 28 anti-fungal compounds with various mechanisms of action.

Publications

- Kenyon-Dean, K., Wang, Z.J., Urbanik, J., Donhauser, K., Hartford, J., Saberian, S., **Sahin, N.**, Bendidi, I., Celik, S., Fay, M., Vera, J.S.R., Haque, I.S., and Kraus, O. (2024). ViTally Consistent: Scaling Biological Representation Learning for Cell Microscopy. *Foundation Models for Science Workshop, 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*. doi: 10.48550/arXiv.2411.02572
- Yeung, C.H.L., **Sahin, N.**, and Andrews, B.J. (2022). Phenomics approaches to understand genetic networks and gene function in yeast. *Biochemical Society Transactions*, **50**(2). doi: 10.1042/BST20210285
- Mattiazzi Ušaj, M.*, **Sahin, N.***, Friesen, H., Pons, C., Ušaj, M., Masinas, M., Shkurin, A., Morris, Q., Boone, C., and Andrews, B.J. (2020). Systematic genetics and single cell image analysis reveals widespread pleiotropy and cell-to-cell variability. *Molecular Systems Biology*, **16**(2), e9243. <https://doi.org/10.15252/msb.20199243> *equal contribution
- Rubanova, Y., Rujan, S., Harrigan, C.F., Li, R., Wintersinger, J., **Sahin, N.**, Deshwar, A., Morris, Q. (2020). Reconstructing evolutionary trajectories of mutation signature activities in cancer using TrackSig. *Nature Communications*, **11**(1):731. doi: 10.1038/s41467-020-14352-7
- Sokolov, A., Ashenden, S., **Sahin, N.**, Lewis, R., Erdem, N., Ozaltan, E., Bender, A., Roth, F.P. and Cokol, M. (2019) Characterizing ABC-Transporter Substrate-Likeness Using a Clean-Slate Genetic Background. *Frontiers in Pharmacology*, **10**:448, doi: 10.3389/fphar.2019.00448
- Grys, B.T., Lo, D.S., **Sahin, N.**, Kraus, O.Z., Morris, Q., Boone, C., and Andrews, B.J. (2017). Machine learning and computer vision approaches for phenotypic profiling. *Journal of Cell Biology* **216**(1), 65–71. <https://doi.org/10.1083/jcb.201610026>
- Chandrasekaran, S., Cokol-Cakmak, M., **Sahin, N.**, Yilancioglu, K., Kazan, H. Collins, J.J. and Cokol, M. (2016). Chemogenomics and orthology-based design of antibiotic combination therapies. *Molecular Systems Biology* **12**(5):872, doi: 10.15252/msb.20156777

Teaching Experience

Winter 2019	University of Toronto – Teaching Assistantship in Computational Biology and Bioinformatics Graduate Course
Fall 2017	University of Toronto – Teaching Assistantship in Computer Science Course CSC120
Fall 2014	Sabanci University – Teaching Assistantship in Cell Biology Course BIO332
2012 – 2015	Sabanci University – Academic Support Program Education Coordinator
2011 – 2012	Sabanci University – Academic Support Program Teaching Assistant

Skills

Programming	Python, Bash, R, C++, Git, GitHub/GitLab
Data Technologies	Scikit-learn, Pandas, Numpy, Matplotlib, Seaborn, Keras, SQL, GCP
Languages	English, Turkish